

5G Core Network Architecture Overview

Section 1. Introduction to Service-Based Architecture

The 3GPP 5G Core (5GC) network adopts a cloud-native Service-Based Architecture (SBA). Control plane Network Functions (NFs) communicate with each other over standardized Service-Based Interfaces (SBI) using HTTP/2 protocol and JavaScript Object Notation (JSON) payloads. The control plane is decoupled from the user plane, allowing independent scaling and distributed deployment.

Section 2. Control Plane Network Functions

The Access and Mobility Management Function (AMF) terminates the Radio Access Network (RAN) control plane on the N2 interface and the Non-Access Stratum (NAS) signaling on the N1 interface. The AMF handles registration management, connection management, reachability management, and mobility management of User Equipment (UE).

The Session Management Function (SMF) handles session establishment, modification, and release. It selects and controls the User Plane Function (UPF) for data routing and allocates IP addresses to user devices.

The Network Repository Function (NRF) provides service registration and discovery, acting as the catalog of all active NF instances.

The Network Slice Selection Function (NSSF) selects the optimal network slice instance for registered users based on subscribed S-NSSAI.

Section 3. User Plane Architecture and UPF Operations

The User Plane Function (UPF) is the primary data path anchor in 5G networks, handling packet routing, forwarding, and Quality of Service (QoS) enforcement.

Key responsibilities of the UPF include:

1. Packet inspection and classification according to Packet Detection Rules (PDR).
2. Traffic forwarding and routing towards external Data Networks (DN) over the N6 interface.
3. Lawful interception and usage reporting for billing through the Charging Function (CHF).
4. Uplink and downlink transport level packet marking using DiffServ (DSCP).
5. Serving as the PDU session anchor point for seamless mobility across gNodeB base stations.

Section 4. Network Slicing and Policy Control

Network slicing partitions physical network resources into dedicated virtual end-to-end networks.

Each slice is characterized by a Slice/Service Type (SST) and an optional Slice Differentiator (SD).

Standard SST values defined by 3GPP include:

- SST 1: Enhanced Mobile Broadband (eMBB) requiring high data rates and spectral efficiency.
- SST 2: Ultra-Reliable Low-Latency Communication (URLLC) with sub-millisecond radio transit time.
- SST 3: Massive Machine Type Communication (mMTC) supporting high connection density for IoT sensors.

The Policy Control Function (PCF) provides unified policy rules to control plane functions and governs dynamic QoS.